

Worked Examples: Compiling Graph Coloring to a Unit-Disk Independent-Set Problem

A step-by-step companion, nothing skipped

Teaching Supplement

May 29, 2026

Abstract

This companion document walks, in full detail, through the transformation that turns a graph-coloring question into the maximum-independent-set (MIS) problem a neutral-atom machine actually solves. It is written for a reader meeting the construction for the first time: every vertex, every edge, and every reasoning step is shown. We treat five graphs of increasing interest — a single edge, a path, a triangle, a square, and a pentagon — and for the smallest one we also carry it all the way onto a unit-disk geometry and solve it by hand. The underlying reduction and pipeline are those of the main paper [1, 2, 3].

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1 The rules of the game

We want to know whether a graph $G = (V, E)$ can be properly colored with k colors: can we assign each vertex one of k colors so that *no edge joins two vertices of the same color*? The hardware cannot answer this directly — it only finds large *independent sets* (sets of mutually non-adjacent vertices) of a *unit-disk graph*. So we translate. The translation has two stages.

Stage 1 — the color-choice graph G' (exact, the heart of everything)

Start from $G = (V, E)$ with $n = |V|$ vertices and $m = |E|$ edges, and a palette of k colors.

1. **Make k copies of every vertex.** The new vertex set is

$$V' = V \times \{1, \dots, k\} = \{(v, i) : v \in V, i \in \{1, \dots, k\}\}.$$

Read (v, i) as the statement “*vertex v takes color i .*” There are $|V'| = nk$ slots.

2. **Choice edges (one color per vertex).** For each original vertex v , join all of its k slots into a clique:

$$\{(v, i), (v, j)\} \text{ for all } i < j.$$

Because an independent set may contain *at most one* vertex of a clique, this forces “ v gets *at most one* color.” This adds $\binom{k}{2}$ edges per vertex.

3. **Conflict edges (adjacent vertices differ).** For each original edge $\{u, v\} \in E$ and each color i , join the two slots of that color:

$$\{(u, i), (v, i)\}.$$

This forbids u and v from *both* taking color i . This adds k edges per original edge.

In total $|E'| = n\binom{k}{2} + mk$.

The payoff. A set $S \subseteq V'$ is independent and has size n *exactly when* it picks one slot per vertex with no conflict — i.e. *exactly when* it is a proper k -coloring. Hence

$$\alpha(G') = n \iff G \text{ is } k\text{-colorable},$$

where α is the size of the largest independent set, and $\alpha(G') \leq n$ always (at most one slot per vertex).

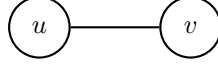
Stage 2 — realize G' as a unit-disk graph the machine can run

G' is still an abstract graph. Stage 2 *places* its vertices (now “atoms”) as points in the plane and picks a radius r so that two atoms are adjacent precisely when they are within distance r . When G' is not directly a unit-disk graph, small *copy* and *crossing* gadgets reproduce the required adjacencies [2, 3]. The machine then finds the maximum-weight independent set of this geometry, and we *decode*: read off, for each original vertex, which color-slot was chosen. We carry this stage out by hand for the single-edge example in Section 7.

How to read the diagrams. For each graph we draw three things: the original graph G ; the color-choice graph G' as a *grid*, with one **column per original vertex** and one **row per color**, so the slot (v, i) sits in column v , row i ; and finally G again, colored by the answer we decode. In the grid, **gray** edges are choice edges (vertical, within a column) and **red** edges are conflict edges (within a row). A **green ring** marks the chosen independent set. Color key:  color 1  color 2  color 3.

2 Example 1 — a single edge K_2

The graph. G has two vertices u, v and one edge. Clearly two colors suffice, and one will not (the two endpoints must differ). Let us watch the machinery confirm $k = 2$.



$$n = 2, m = 1, k = 2.$$

Step 1 — the slots. Two vertices $\times$ two colors give $|V'| = nk = 4$ slots:

$$(u, 1), (u, 2), (v, 1), (v, 2).$$

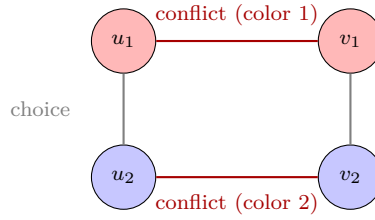
Step 2 — choice edges. One clique per vertex; with $k = 2$ a clique is a single edge ($\binom{2}{2} = 1$ edge each):

$$\{(u, 1), (u, 2)\}, \quad \{(v, 1), (v, 2)\}.$$

Step 3 — conflict edges. The single edge $\{u, v\}$ contributes one conflict edge per color:

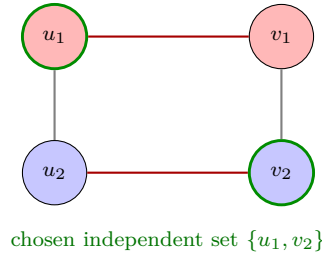
$$\text{color 1: } \{(u, 1), (v, 1)\}, \quad \text{color 2: } \{(u, 2), (v, 2)\}.$$

Step 4 — tally. $|V'| = 4$ and $|E'| = n\binom{k}{2} + mk = 2 \cdot 1 + 1 \cdot 2 = 4$. Here is G' :



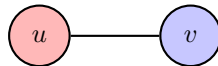
Notice G' is a 4-cycle $u_1 - u_2 - v_2 - v_1 - u_1$.

Step 5 — find the largest independent set. We may take at most one slot from each column (the choice edge forbids both). Try one per column: choosing $(u, 1)$ rules out $(v, 1)$ (conflict, color 1), leaving $(v, 2)$. Check the pair $S = \{(u, 1), (v, 2)\}$: $(u, 1)$'s neighbors are $(u, 2)$ and $(v, 1)$ — neither is $(v, 2)$ — so S is independent, and $|S| = 2 = n$.



Step 6 — the verdict. $\alpha(G') = 2 = n$, so by the boxed rule G is 2-colorable.

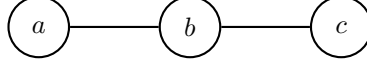
Step 7 — decode. The chosen slots say $u \mapsto \text{color 1}$, $v \mapsto \text{color 2}$. Restoring this onto G :



A proper 2-coloring, exactly as expected.

3 Example 2 — a path P_3

The graph. Three vertices in a line, $a - b - c$. It is bipartite, so two colors suffice. $n = 3$, $m = 2$, $k = 2$.



Step 1 — slots. $|V'| = nk = 6$: $(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)$.

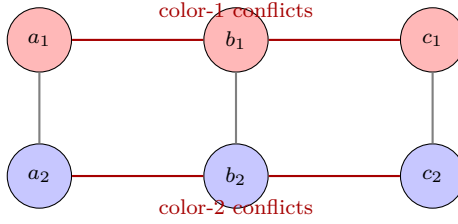
Step 2 — choice edges. One per column: $\{(a, 1), (a, 2)\}, \{(b, 1), (b, 2)\}, \{(c, 1), (c, 2)\}$.

Step 3 — conflict edges. Two original edges $\{a, b\}$ and $\{b, c\}$, each duplicated over the two colors:

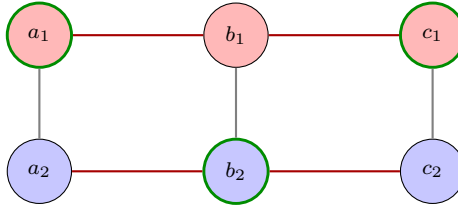
from ab : $\{(a, 1), (b, 1)\}, \{(a, 2), (b, 2)\}$; from bc : $\{(b, 1), (c, 1)\}, \{(b, 2), (c, 2)\}$.

There is *no* conflict edge between columns a and c , because a and c are *not* adjacent in G — a point we exploit in a moment.

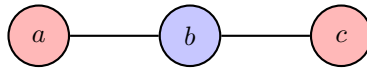
Step 4 — tally. $|E'| = 3 \cdot 1 + 2 \cdot 2 = 7$. Here is G' :



Step 5 — largest independent set. The standard proper coloring is $a = 1$, $b = 2$, $c = 1$. The matching slots are $S = \{(a, 1), (b, 2), (c, 1)\}$. Check: $(a, 1)$ touches only $(a, 2), (b, 1)$; $(b, 2)$ touches only $(b, 1), (a, 2), (c, 2)$; $(c, 1)$ touches only $(c, 2), (b, 1)$. None of these hit another member of S — in particular $(a, 1)$ and $(c, 1)$ share row 1 but are *not* joined, since a, c are non-adjacent. So S is independent and $|S| = 3 = n$.

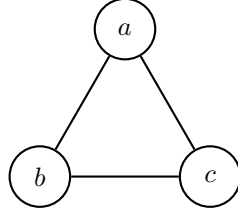


Steps 6–7 — verdict and decode. $\alpha(G') = 3 = n$: G is 2-colorable, with $a = 1$, $b = 2$, $c = 1$.



4 Example 3 — a triangle K_3 : when k is too small

The graph. Three mutually adjacent vertices a, b, c . Every pair is an edge, so all three must receive *different* colors: $\chi(K_3) = 3$. We deliberately first *try* $k = 2$ *and watch it fail*, then succeed with $k = 3$. $n = 3$, $m = 3$.

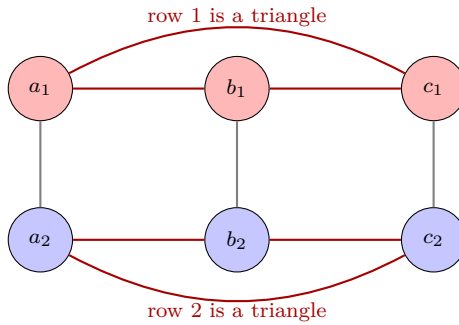


4.1 Attempt with $k = 2$ (must fail)

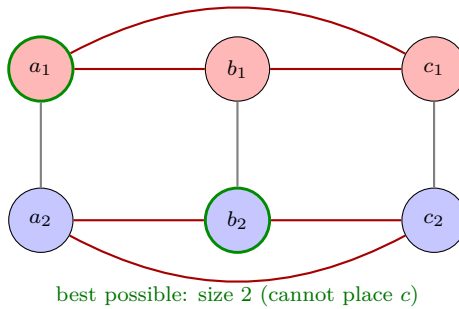
Steps 1–3. Slots $\{(a, i), (b, i), (c, i) : i = 1, 2\}$, $|V'| = 6$. Choice edges per column as usual. Now *every* original edge is present, so for each color i the three slots $(a, i), (b, i), (c, i)$ are pairwise joined — each row is itself a triangle:

color 1: $\{(a, 1), (b, 1)\}, \{(b, 1), (c, 1)\}, \{(a, 1), (c, 1)\}$; color 2: the same with subscript 2.

$$|E'| = 3 \cdot 1 + 3 \cdot 2 = 9.$$



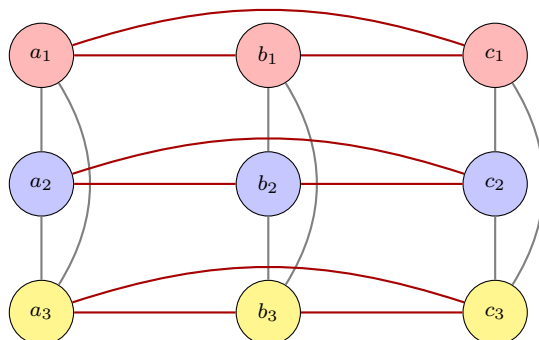
Step 5 — why size 3 is impossible. An independent set takes at most one slot per column, so at most 3 slots, one in each of columns a, b, c . Those three live in only *two* rows, so by the pigeonhole principle two of them share a row. But within a row *every* pair is a conflict edge — so those two are adjacent, contradicting independence. Hence no independent set of size 3 exists. The best we can do is size 2, e.g. $\{(a, 1), (b, 2)\}$:



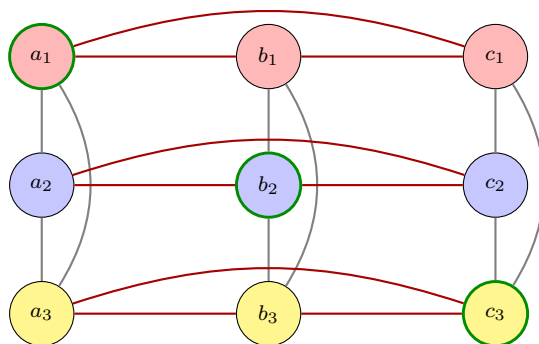
Step 6 — verdict. $\alpha(G') = 2 < 3 = n$, so G is *not* 2-colorable. The reduction reports failure honestly: there simply is no size- n independent set to find.

4.2 Attempt with $k = 3$ (succeeds)

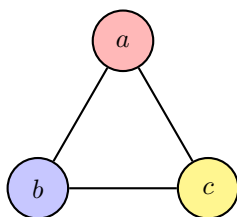
Now each vertex has three slots, $|V'| = nk = 9$. Choice edges make each column a triangle; conflict edges make each of the three rows a triangle. $|E'| = 3\binom{3}{2} + 3 \cdot 3 = 9 + 9 = 18$.



Step 5 — a size-3 set now exists. Put each vertex in a *different* row: $S = \{(a, 1), (b, 2), (c, 3)\}$. Members lie in distinct columns (no choice edge) *and* distinct rows (conflict edges only ever join slots in the *same* row), so no edge joins any two of them. S is independent and $|S| = 3 = n$.



Steps 6–7 — verdict and decode. $\alpha(G') = 3 = n$: K_3 is 3-colorable, decoded as $a = 1$, $b = 2$, $c = 3$ — three different colors, as it must be.

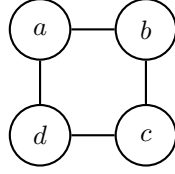


Lesson

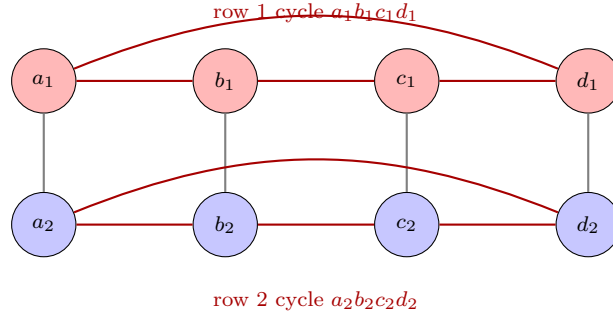
The same graph fails at $k = 2$ and succeeds at $k = 3$. To find the *chromatic number* we simply try increasing k and report the first that yields $\alpha(G') = n$. For K_3 that first success is $k = 3$, so $\chi(K_3) = 3$.

5 Example 4 — a square C_4 (even cycle)

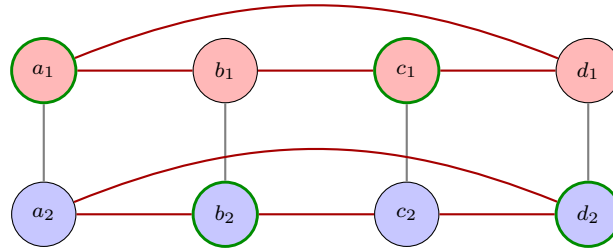
The graph. A 4-cycle $a - b - c - d - a$. Even cycles are bipartite, so two colors suffice. $n = 4$, $m = 4$, $k = 2$.



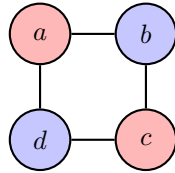
Steps 1–3. $|V'| = 8$. Choice edges per column. The four original edges ab, bc, cd, da each appear in both colors. So in each row the conflict edges trace the same 4-cycle on the columns: $a_i - b_i - c_i - d_i - a_i$. Crucially there is *no* edge between columns a and c , nor between b and d (those pairs are non-adjacent in G). $|E'| = 4 \cdot 1 + 4 \cdot 2 = 12$.



Step 5 — largest independent set. The proper 2-coloring alternates around the square: $a = 1, b = 2, c = 1, d = 2$. The slots are $S = \{(a, 1), (b, 2), (c, 1), (d, 2)\}$. The two row-1 members $(a, 1), (c, 1)$ are *not* adjacent (no $a-c$ edge), and likewise $(b, 2), (d, 2)$ are not adjacent; across rows there are no conflicts. So S is independent, $|S| = 4 = n$.

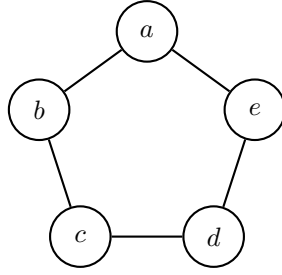


Steps 6–7 — verdict and decode. $\alpha(G') = 4 = n$: C_4 is 2-colorable, $a = c = 1, b = d = 2$.



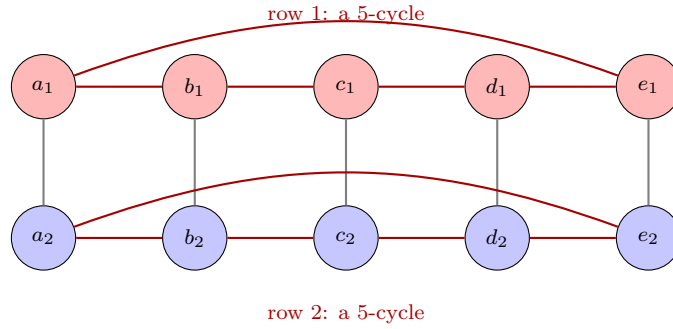
6 Example 5 — a pentagon C_5 (odd cycle)

The graph. A 5-cycle $a - b - c - d - e - a$. Odd cycles are the smallest graphs that are *not* bipartite: two colors fail, three succeed, so $\chi(C_5) = 3$. Contrast this with the even cycle C_4 above — the parity of the cycle is the whole story. $n = 5, m = 5$.

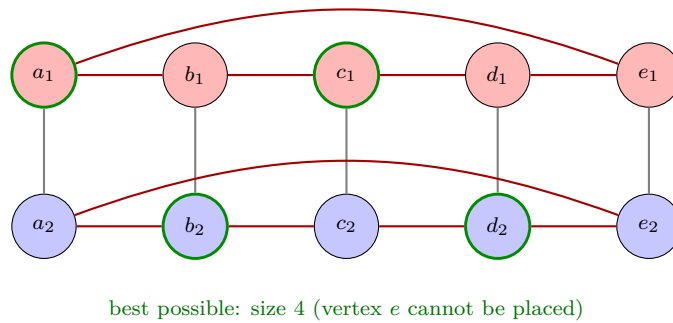


6.1 Attempt with $k = 2$ (must fail)

Steps 1–3. $|V'| = 10$. Choice edge per column. Each of the five edges ab, bc, cd, de, ea appears in both rows, so each row is a 5-cycle on the columns. $|E'| = 5 \cdot 1 + 5 \cdot 2 = 15$.



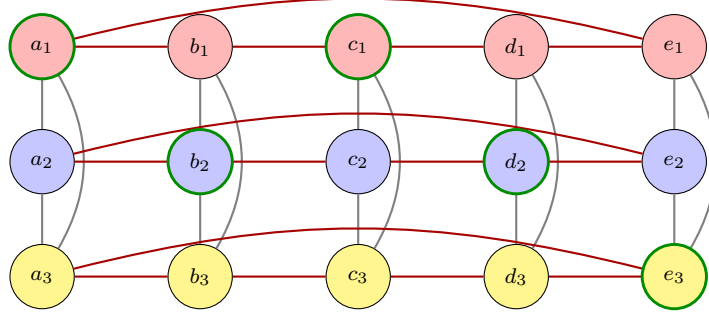
Step 5 — why it fails. An independent set picks one slot from each of the five columns, placing five members into only *two* rows. Walk around the cycle a, b, c, d, e : a 2-coloring would have to alternate $1, 2, 1, 2, \dots$, but after five steps we return to a with a clash — an odd cycle cannot be 2-colored. Concretely, any one-per-column choice forces two *adjacent* vertices into the same row, and a same-row adjacent pair is a conflict edge. So no independent set reaches size 5; the best is size 4, e.g. color a, b, c, d as $1, 2, 1, 2$ and leave e out (both e_1 and e_2 clash — e_1 with a_1 , and e_2 with d_2):



Step 6 — verdict. $\alpha(G') = 4 < 5 = n$: C_5 is not 2-colorable.

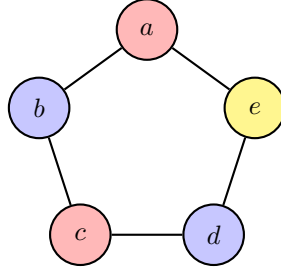
6.2 Attempt with $k = 3$ (succeeds)

Add a third row. $|V'| = 15$, $|E'| = 5\binom{3}{2} + 5 \cdot 3 = 15 + 15 = 30$. A proper 3-coloring of the pentagon is $a = 1, b = 2, c = 1, d = 2, e = 3$: the new third color rescues the one vertex that the alternation could not place. The matching independent set is $S = \{(a, 1), (b, 2), (c, 1), (d, 2), (e, 3)\}$.



Every chosen slot sits in a distinct column, and adjacent cycle-vertices land in distinct rows, so S has no internal edge: it is independent of size $5 = n$.

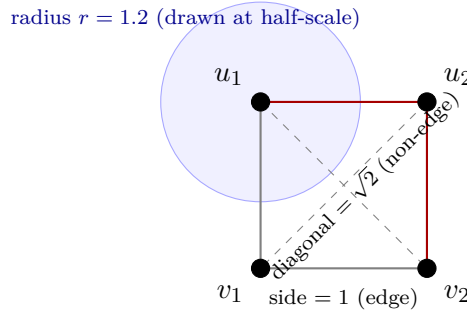
Steps 6–7 — verdict and decode. $\alpha(G') = 5 = n$: C_5 is 3-colorable, decoded as $a = 1, b = 2, c = 1, d = 2, e = 3$. Since $k = 2$ failed, $\chi(C_5) = 3$.



7 Carrying Example 1 onto the hardware geometry

Stages 1 finished with an *abstract* graph G' . The machine, however, needs *points in the plane*: atoms are adjacent exactly when they sit within the blockade radius r . We now finish the single-edge example (Section 2) by hand, because there G' is small enough to place directly.

Recall G' was the 4-cycle $u_1 - u_2 - v_2 - v_1 - u_1$. Its *edges* are the four sides of the cycle; its *non-edges* are the two “diagonals” $\{u_1, v_2\}$ and $\{u_2, v_1\}$. To realize this as a unit-disk graph we place the four atoms at the corners of a unit square in cycle order, so that each *side* (an edge of G') has length 1 and each *diagonal* (a non-edge) has length $\sqrt{2} \approx 1.41$. Choosing any radius with $1 \leq r < \sqrt{2}$, say $r = 1.2$, reproduces G' exactly: sides fall inside the blockade, diagonals fall outside.



Solve the MWIS by hand. With all weights equal, the maximum independent set of a 4-cycle has size 2: pick two atoms that are *not* joined — a diagonal pair. Take $\{u_1, v_2\}$: they sit at distance $\sqrt{2} > r$, so they are non-adjacent and may be excited together. The many-body ground state of the atom array settles on exactly such a pair.

Decode. Read each chosen atom as “vertex takes that color”: u_1 means $u \mapsto 1$, and v_2 means $v \mapsto 2$. This is the same proper coloring we found abstractly — now produced by geometry and (in the real device) by physics. The other ground state $\{u_2, v_1\}$ decodes to the mirror coloring $u \mapsto 2, v \mapsto 1$; both are correct.

What happens for the larger examples

For P_3 , C_4 , C_5 , and the $k = 3$ cases, the color-choice graph G' is no longer a plain 4-cycle and usually is *not* itself a unit-disk graph. Stage 2 then inserts *copy* gadgets (short chains of atoms that carry an occupation signal from one place to another) and *crossing* gadgets (that let two such signals pass without interacting), after which the whole instance is a weighted unit-disk graph whose maximum-weight independent set still decodes to a coloring [2, 3]. The bookkeeping grows, but every step is the same kind of mechanical, exactly-reversible move shown above — which is why the main paper treats this stage as a once-proved, correctness-preserving black box and spends its real effort on *placing* the atoms within the device’s size and spacing limits.

8 What the five examples taught us

Graph	n	m	k	$ V' = nk$	$\alpha(G')$	verdict
single edge K_2	2	1	2	4	2	2-colorable
path P_3	3	2	2	6	3	2-colorable
triangle K_3	3	3	2	6	2	<i>not</i> 2-colorable
triangle K_3	3	3	3	9	3	3-colorable $\Rightarrow \chi = 3$
square C_4	4	4	2	8	4	2-colorable
pentagon C_5	5	5	2	10	4	<i>not</i> 2-colorable
pentagon C_5	5	5	3	15	5	3-colorable $\Rightarrow \chi = 3$

Table 1: Every row obeys the single rule $\alpha(G') = n \iff k\text{-colorable}$. A “not colorable” verdict shows up cleanly as $\alpha(G') < n$.

Three ideas recur:

- **One slot per vertex, never two:** the choice clique caps every independent set at n , so reaching exactly n is the whole game.
- **Conflicts live inside a single color:** two same-colored adjacent vertices are the *only* way a one-per-vertex pick can fail, which is why each row carries a copy of G ’s edges.
- **Too few colors = pigeonhole:** K_3 at $k = 2$ and C_5 at $k = 2$ fail for the same reason — more vertices to seat than rows to seat them in, forcing an in-row clash. Adding a row (a color) is exactly what relieves the pressure.

References

- [1] R. M. Karp. *Reducibility among combinatorial problems*. In *Complexity of Computer Computations*, 85–103, Springer, 1972.
- [2] H. Pichler, S.-T. Wang, L. Zhou, S. Choi, M. D. Lukin. *Quantum optimization for maximum independent set using Rydberg atom arrays*. arXiv:1808.10816, 2018.

- [3] M.-T. Nguyen, J.-G. Liu, J. Wurtz, M. D. Lukin, S.-T. Wang, H. Pichler. *Quantum optimization with arbitrary connectivity using Rydberg atom arrays*. PRX Quantum 4, 010316, 2023.